moving through. Just like a curved fiber-optic cable can make the light inside it turn a corner, a massive object bending space can cause light to curve around it. Galaxies and clusters of galaxies become distorting magnifying glasses for the objects behind them. Some of our most compelling evidence for dark matter comes from finding that this "gravitational lensing" effect is stronger than can be accounted for by the mass of the stuff we can actually see—some of the mass is due to something invisible. It turns out there's a *lot* of dark matter out there. The first attempts to weigh up the matter in the universe by looking at just the visible stuff gave us a wildly inaccurate accounting. Not long after Vera Rubin's studies, it became clear that the vast majority of the matter in the universe is dark.

But even when dark matter was properly accounted for, it was difficult to determine whether the density of matter in space was on one side or the other of the "critical density" that defined the border between a recollapsing universe and an eternally expanding one. Determining the contents of the universe was only one part of the problem; the other part was figuring out exactly how fast the universe is expanding, or, alternatively, how the expansion has changed over cosmic time. This, it turns out, is no mean feat.

In order to get a good measurement of the cosmic expansion rate for a reasonable fraction of the history of the universe, you need to survey a huge number of galaxies, at a range of distances. Then, for each galaxy, you need to work out two things: its speed and its actual physical distance from us. Astronomers worked out the *local* expansion rate with the Hubble-Lemaître Law back in 1929 (though the exact number for the proportionality was argued about for decades after, and is still a point of some controversy). But to answer the Big Crunch question, we need to know the expansion rate across a huge swath of cosmic time, which means a huge distance in space. That's not so much of a problem for the galaxy speed part of the equation, since this can be determined with redshift measurements, which are, generally, reasonably straightforward. Measuring *distance* accurately over billions of light-years, however, is a lot harder.

Studying the distances and speeds of galaxies using images from photographic plates in the late 1960s led astronomers to state with increasing confidence, despite quite a lot of remaining uncertainty, that we were, in fact, fated to collapse. This prompted a few astronomers to write